

Clostridium Difficile Diarrhea

Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency

Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

DRUG INTERACTIONS

Digoxin

Simultaneous treatment with omeprazole and digoxin in healthy subjects lead to a 10% increase in the bioavailability of digoxin as a consequence of the increased intragastric pH.

Diazepam, Phenytoin and Warfarin

Omeprazole inhibits the metabolism of drugs metabolised by the hepatic cytochrome P450 enzyme system and may increase plasma concentrations of diazepam, phenytoin, and warfarin.

Aminopyrine and Antipyrine

Administration of 60 mg daily for 2 weeks caused a 10-20% prolongation of aminopyrine and antipyrine half-life while half this dose had no significant effect. In most animal models, the effects of omeprazole on barbiturate sleeping times and antipyrine clearance are less marked than those caused by cimetidine.

Theophylline, Propranolol and 7 ethoxycoumarin de-ethylase

Omeprazole does not interfere with theophylline or propranolol metabolism. In one study, however, omeprazole was a more potent inhibitor of 7 ethoxycoumarin de-ethylase activity than cimetidine.

Metoprolol, Lidocaine, Quinidine, Amoxycillin or Antacids

There is no evidence of interaction with theophylline, propranolol, metoprolol, lidocaine, quinidine, amoxycillin or antacids, but interaction with other drugs also metabolised via cytochrome P-450 system cannot be excluded. Combinations containing any of the following medications, depending on the amount present, may also interact with this medication.

Ampicillin esters, Iron salts or Ketoconazole

(Omeprazole may increase gastrointestinal pH; concurrent use with omeprazole may result in a reduction in absorption of ampicillin esters, iron salts, or ketoconazole)

Bone marrow depressants

(Concurrent use of omeprazole with these medications may increase the leukopenic and/or thrombocytopenic effects of both these medications; if concurrent use is required, close observations for toxic effects should be considered)

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:

Omilock Capsules are recommended to be given in the morning and swallowed whole with liquid.

For patients with swallowing difficulties the capsule might be opened and the contents swallowed or suspended in a slightly acidic fluid e.g juice, soured milk, or non-carbonated water. The dispersion should be taken immediately or within 30 minutes. Alternatively patients can suck the capsule and swallow the pellets with liquid. The pellets must not be chewed or crushed.

For duodenal and benign gastric ulcers, the usual dose is 20 mg Omilock once daily. Symptom resolution is rapid. The majority of patients with duodenal ulcer are healed after 4 weeks. The majority of patients with benign gastric ulcer are healed after 8 weeks. In severe cases, the dose may be increased to 40 mg Omilock once daily. Long term therapy for the prevention of relapse in patients with poorly responsive peptic ulcer is recommended at a dosage of 20-mg Omilock Capsule once daily.

For reflux oesophagitis, the usual dosage is 20 mg Omilock 20 mg once daily. The majority patients are healed after 4 weeks. For those patients not fully healed after the initial course, healing usually occurs during a further 4-8 weeks treatment. Omilock Capsule has also been used in a dosage of 40 mg once daily in patients with reflux oesophagitis refractory to other therapy. Healing usually occurred within 8 weeks. Patients can be continued to a dosage of 20 mg once daily.

Zollinger-Ellison syndrome: The recommended initial dosage is 60-mg Omilock Capsule once daily. The dosage should be adjusted individually and treatment continued as long as clinically indicated. >90% of patients with severe disease and inadequate response to other therapies have been effectively controlled on doses of 20-120 mg daily. With doses >80 mg daily, the dose should be divided and given twice daily.

Helicobacter pylori-associated peptic ulcer disease:

The usual dose is Omeprazole Capsule 40 mg daily with oral amoxycillin 1.5 g daily (given as 750 mg twice daily) for 2 weeks. In clinical studies, daily dose of 1.5 to 2 g of amoxycillin have been used.

Impaired renal function:

Dose adjustment is not required in patients with impaired renal function.

Impaired hepatic function:

As bioavailability and plasma half-life of Omeprazole are increased in patients with impaired hepatic function, a daily dose of 10-20 mg may be sufficient.

SYMPTOMS AND TREATMENT FOR OVERDOSE, AND ANTIDOTE(S):

None has been reported; Single oral doses of up to 160 mg and total oral doses of up to 300 mg/day have been given without adverse effects.

Symptoms / Clinical Effects of overdose

The following effects have been selected on the basis of their potential clinical significance: Blurred vision; confusion; diaphoresis (increased sweating); drowsiness; dryness of mouth; flushing; headache; nausea; tachycardia (fast or irregular heartbeat)

Treatment

Since there is no specific antidote for overdose with omeprazole, treatment should be symptomatic and supportive. Due to extensive protein binding, omeprazole is not readily dialyzable. Patients in whom intentional overdose is confirmed or suspected should be referred for psychiatric consultation.

PACKING / PACK SIZES:

Blister packs of 2 x 7's/Box.

STORAGE CONDITIONS, USER

INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:

Keep all medicines out of reach of children.
Store in a dry place, below 30°C.
Protect from light.

SHELF LIFE:

3 years.

Manufacturer and Product Registration Holder:

SM PHARMACEUTICALS SDN BHD
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SM PHARMACEUTICALS SDN. BHD.

OMILOCK (OMEPRAZOLE)

CAPSULE 20 MG

DESCRIPTION:

20 mg: Caramel/Pink coloured hard gelatin capsule, size "2" containing off-white granules.

COMPOSITION:

Each capsule contains: Omeprazole 20 mg

Pharmacodynamics

Omeprazole is a selective and irreversible proton pump inhibitor. Omeprazole suppresses gastric acid secretion by specific inhibition of the hydrogen potassium adenosinetriphosphatase (H⁺, K⁺-ATPase) enzyme system found at the secretory surface of parietal cells. It inhibits the final transport of hydrogen ions (via exchange with potassium ions) into the gastric lumen. Since the H⁺, K⁺-ATPase enzyme system is regarded as the acid (proton) pump of the gastric mucosa, omeprazole is known as a gastric acid pump inhibitor. The inhibitory effect is dose-related. Omeprazole inhibits both basal and stimulated acid secretion irrespective of the stimulus. Omeprazole does not have anticholinergic or histamine H₂-receptor antagonist properties.

During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours. Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

Pharmacokinetics:

Absorption:

Rapid.

Absolute bioavailability is about 30 to 40% at doses of twenty to forty milligrams, due in large part to presystemic metabolism. Bioavailability in patients with chronic hepatic disease is about 100%, reflecting decreased first-pass effect. Bioavailability in healthy elderly volunteers was 76%, as compared with 58% in young volunteers.

Distribution

Distributed in tissue, particularly gastric parietal cells.

Protein binding

Very high, approximately 95% bound to albumin and alpha1-acid glycoprotein.

Biotransformation

Hepatic, extensive. Omeprazole is subject to saturable, first-pass metabolism, and is completely and rapidly metabolized by the hepatic P450 (CYP) enzyme system.

Half-life

Plasma

Normal hepatic function 30 minutes to 1 hour.

Chronic hepatic disease 3 hours.

Onset of action

Within one hour.

Time to peak concentration

Within 30 minutes to 3.5 hours.

Time to peak effect

Within 2 hours.

Duration of action

Up to 72 hours or more (96 hours required for full restoration of acid production).

Elimination

Renal 70 to 77%. No unchanged omeprazole was detected in urine.

Fecal 18 to 23%.

In dialysis Not readily dialyzable, because of extensive protein binding.

Pharmacogenetics

Pharmacokinetic studies in Asian subjects receiving single 20-mg doses of omeprazole showed an

approximately fourfold increase in the area under the plasma concentration-time curve (AUC) as compared to Caucasian subjects. Dosage adjustments should be considered for Asian patients, especially for prophylaxis of recurrence of erosive esophagitis.

INDICATIONS:

1. Treatment of duodenal and benign gastric ulcers including those complicating NSAID therapy.
2. *Helicobacter pylori* associated peptic ulcer.
3. Treatment of reflux oesophagitis.
4. Zollinger – Ellison syndrome.

CONTRAINDICATIONS:

There are no known contraindications to the use of Omilock Capsule.

Risk-benefit should be considered when the following medical problem exist:

Sensitivity to Omeprazole.

SIDE EFFECTS / ADVERSE REACTIONS:

Omeprazole capsule is well tolerated and adverse reactions have generally been mild and reversible. The following side / adverse effects have been selected on the basis of their potential clinical significance (possible signs and symptoms in parentheses where appropriate) not necessarily inclusive:

Those indicating need for medical attention

Incidence rare

Generalized skin reactions, including toxic epidermal necrolysis (blisters; chills; fever; general feeling of discomfort or illness; muscle aches; red or irritated eyes; redness, itching, burning, or peeling of skin; sore throat; sores or ulcers on lips or in mouth) sometimes fatal; **Stevens-Johnson syndrome** (bleeding or crusting sores on lips; chills; fever; muscle, cramps; pain; skin rash or itching; sore throat; sores; ulcers, or white spots on lips, in mouth, or on genitals); **or erythema multiforme** (blisters on palms of hands and soles of feet; fever, general feeling of discomfort or illness; joint pain; redness of skin); **hematologic abnormalities, specifically anemia** (unusual tiredness or weakness); **agranulocytosis** (chills; fever; sore throat; unusual tiredness or weakness) sometimes fatal; **haemolytic anemia** (back, leg, or stomach pain; loss of appetite; unusual tiredness or weakness); **leukocytosis** (sore throat and fever); **neutropenia** (continuing ulcers or sores in mouth); **pancytopenia or thrombocytopenia** (unusual bleeding or bruising); **hematuria** (bloody urine); **proteinuria** (cloudy urine); **urinary tract infection** (bloody or cloudy urine; difficult, burning, or painful urination; frequent urge to urinate)

Those indicating need for medical attention only if they continue or are bothersome

Incidence more frequent

Abdominal pain or colic

Incidence less frequent

Asthenia (muscle pain; unusual tiredness); **back pain, central nervous system (CNS) disturbances, specifically dizziness, headache, somnolence** (unusual drowsiness); or unusual tiredness; chest pain; gastrointestinal disturbances, specifically acid regurgitation (heartburn); **constipation; diarrhoea or loose stools; flatulence** (gas); **or nausea and vomiting; skin rash or itching.**

Also reported are somnolence, insomnia and vertigo.

In isolated cases, the following have been reported: Blurred vision, taste disturbance, peripheral oedema, increased sweating, gynaecomastia, bronchospasm, encephalopathy in patients with preexisting severe liver disease, hepatitis with or without jaundice, rarely interstitial nephritis and hepatic failure. Increase in liver enzymes have been observed.

Subacute Cutaneous Lupus Erythematosus (SCLE)

Skin and subcutaneous tissue disorders Frequency 'not known': Subacute cutaneous lupus erythematosus

Interstitial Nephritis

Renal and urinary disorders: Interstitial nephritis

Hypomagnesaemia

Metabolism and nutritional disorders Frequency "not known": hypomagnesaemia.

Fracture

Musculoskeletal disorders Frequency "uncommon": Fracture of the hip, wrist or spine.

Clostridium Difficile Diarrhea

Infections & infestations: Clostridium difficile associated diarrhea.

Fundic Gland Polyps (Benign)

Gastrointestinal disorders Frequency "common": Fundic gland polyps (benign)

Vitamin B12 Deficiency

Metabolic/Nutritional: Vitamin B12 deficiency

Gastrointestinal disorders

Microscopic colitis : Frequency 'not known'

PRECAUTIONS / WARNINGS:

Before giving omeprazole to patients with gastric ulcers the possibility of malignancy should be excluded since omeprazole may mask symptoms and delay diagnosis.

Pregnancy / Reproduction

Fertility In a rat fertility and general reproductive performance test, omeprazole, in a dose 35 to 345 times the human dose, was not toxic or deleterious to the reproductive performance of parental animals.

Pregnancy : Adequate and well-controlled studies in humans have not been done. Sporadic instances of congenital abnormalities in infants born to women who received omeprazole during pregnancy have been reported.

Studies in pregnant rats did not show omeprazole to have any teratogenic potential at doses 345 times the human dose. Omeprazole produced dose-related increases in embryo-lethality, fetal resorptions, and pregnancy disruptions in rabbits receiving 17 to 172 times the human dose. In rats, dose-related embryo/fetal toxicity and postnatal developmental toxicity were observed in offspring resulting from parents treated with 35 to 345 times the human dose.

FDA Pregnancy Category C.

Breast-feeding

It is not known whether omeprazole is distributed into human breast milk. However, because omeprazole has been shown to cause tumorigenic and carcinogenic effects in animals, risk-benefit must be considered.

Neonates

There are no data and the drug is not recommended for neonates.

Paediatrics

Appropriate studies on the relationship of age to the effects of omeprazole have not been performed in the paediatric population and the drug is not recommended. Safety and efficacy have not been established.

Geriatrics

No information is available on the relationship of age to the effects of omeprazole in geriatric patients. However, a somewhat decreased rate of elimination and an increased bioavailability are more likely to occur in geriatric patients taking omeprazole.

Toxicology

General tolerance is good with long-term use in doses of up to 138 mg/kg in dogs and 414 mg/kg in rats and mice.

Gastric mucosal cell hyperplasia has been observed in animals after 3 months high-dose therapy and these changes are prevented by antrectomy. Carcinoid tumours and enterochromaffin-like (ECL) cell hyperplasia have been reported to occur in rats treated for two years with doses ranging from 14 to 140 mg/kg. The ECL cell changes were dose-related and female rats were particularly susceptible. ECL cell hyperplasia is reversible on cessation of high-dose omeprazole. The development of ECL cell hyperplasia and carcinoids is thought to be related to hypergastrinaemia produced by prolonged hypochlorhydria. Support for this hypothesis is strong.

Plasma gastrin levels in rats receiving omeprazole correlate directly with acid secretory inhibition. Reversible, coordinated gastric mucosal hypertrophy is related to hypergastrinaemia and does not occur in antrectomized animals. It is not clear, however, why ECL cell growth outstrips that of other mucosal constituents with subsequent development of autonomy. In man, circulating levels of gastrin rise during omeprazole treatment. However, after four week omeprazole therapy gastrin levels in man are generally considerably lower than levels in treated rodents, in the Zollinger-Ellison syndrome or in pernicious anaemia. Mucosal ECL cell morphology appears unchanged in man following short or long-term use. Thus the relevance of gastric mucosal changes in rats and dogs to man is unclear. Drug regulatory bodies in Europe, North America and Australia have accepted this hypothesis.

Interference with laboratory tests

Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patient(s) are due to have a test on Chromogranin A level, [product name] treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see section Pharmacodynamic). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

Regular Surveillance

Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Subacute Cutaneous Lupus Erythematosus (SCLE)

Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Omilock. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Hypomagnesaemia

Severe hypomagnesaemia has been reported in patients treated with PPI like [product name] for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fracture

Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognised risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10-40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.